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6. Scenario: You are a cashier at a grocery store and need to calculate the total cost of a

customer' s purchase, including applicable discounts and taxes. You have the item prices and

quantities in separate lists, and the discount and tax rates are given as percentages. Your task is to

calculate the total cost for the customer.

Question: Use arithmetic operations to calculate the total cost of a customer's purchase,

including discounts and taxes, given the item prices, quantities, discount rate, and tax rate?

Code;

```
prices = [100, 200, 150]

quantities = [2, 1, 3]

discount_rate = 10

tax_rate = 5


subtotal = sum(prices[i] * quantities[i] for i in range(len(prices)))


discount = subtotal * discount_rate / 100

amount = subtotal - discount


tax = amount * tax_rate / 100

total_cost = amount + tax


print("Subtotal:", subtotal)

print("Discount:", discount)
```

print("Tax:", tax)

print("Total Cost:", total_cost)

input;

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print("Subtotal:", subtotal)
print("Discount:", discount)
print("Tax:", tax)
print("Total Cost:", total_cost)
```

Output;

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```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on w
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/Jashw/exxpp6 fods.py ==========
Subtotal: 850
Discount: 85.0
Tax: 38.25
Total Cost: 803.25
```